

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	(add) {0.004 g/4mg} in 20 cm³ of water	Accept equivalent correct answers e.g. 4mg in 0.02dm ³ Or 200mg in 1dm ³ and then remove 20cm ³	Exp (1)

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	0.93	Accept 0.933/0.9 Ignore 0.93333/0.93	(1) Exp

Question Number	Answer	Additional Guidance	Mark
2(a)(iii)	56.99	ECF from part ii Allow 55.55 to 57.127/57.13 Max 3 decimal places	Exp (1)
Question Number	Answer	Additional Guidance	Mark
2(a)(iv)	An explanation including two of the following: • facilitated diffusion (down a concentration gradient) (1) • (concentration) gradient reduces over time (1)	Accept description e.g. channel proteins Accept description of gradient and time refs using 1 - 2 hrs and 4 - 5hrs	Exp (2)

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	Abiotic • temperature • pH Biotic • age / sex of toad • {part of body providing skin sample/thickness of skin}	Ignore other abiotic factors Accept gender Accept disease If biotic and abiotic factor stated on the same line they negate each other	Exp (2)

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	An answer that includes two of the following: • variable with suitable control method described (1) • results are not valid / description of expected effect on the dependent variable (1)	Accept AC / incubator/thermostatically controlled waterbath Accept stated directional answer	Exp (2)

(Total for Question 2 = 9 marks)

